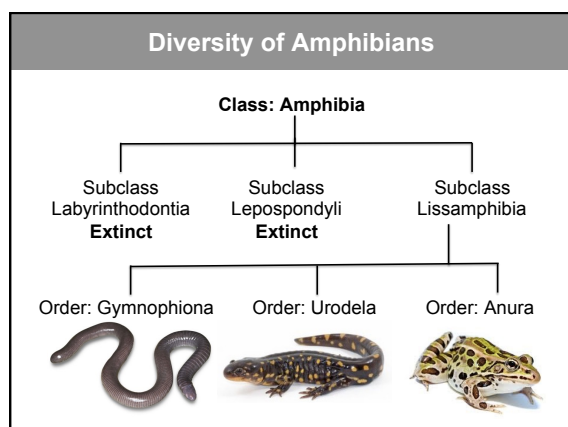




Diversity of Amphibians

Order: Gymnophiona/Apoda

- *Gymnos* - naked, *opineos* - snake
- 215 species
- Origins – 200 MYA (Triassic period)
- Distribution: Africa, South Asia, SE Asia, Tropical America
- **Body- long slender and limbless**
- Length from 30 cm to 1.5 m
- **All are burrowing or aquatic**
- Commonly called **caecilians**



Diversity of Amphibians

Order: Gymnophiona/Apoda

- Respiration
 Adults - Lungs, skin
 Larvae – external gills
- **Some bear small embedded scales in the skin**



Diversity of Amphibians

Order: Gymnophiona/Apoda

- **Eyes are small, and most species are totally blind**
- **Special sensory tentacles occur on the snout**
- Food – **mostly worms and small invertebrates**



Diversity of Amphibians

Order: Gymnophiona/Apoda

Reproduction

- Fertilization internal, males bear a protrusible copulatory organ - **Phallodeum**
- Eggs usually are deposited in moist ground near water
- In some, larvae are aquatic
- Some show direct development: complete larval development within the egg



Diversity of Amphibians

Order: Urodela/Caudata

- 'Having a tail'
- Origins - about 200 Mya
- Consists of tailed amphibians
- Nearly 765 species
- Commonly called **Salamanders or Newts**
- Found in temperate regions of Europe, Asia, America and Tropical America
- Usually bears four limbs



Diversity of Amphibians

Order: Urodela/Caudata

- **Limbs set at right angles to the body**
- Forelimbs and hind limbs are roughly equal size
- Some aquatic, arboreal and burrowing forms- limbs are very small or absent
- Size – Length 15 cm to 1.5 m



Diversity of Amphibians

Order: Urodela/Caudata

- Diet- **carnivorous** both as larvae and adults
- Prey- worms, small arthropods, and small mollusks

Respiration

- Generally terrestrial forms use **lungs** as well as the **skin** (cutaneous respiration)
- Aquatic larvae use **external gills**
- Completely aquatic forms retain gills
- Some terrestrial forms have lost lungs, entirely depend on **cutaneous respiration**

Diversity of Amphibians

Order: Urodela/Caudata

Reproduction

- Fertilization – **internal in majority**
- Generally an aquatic larval stage is present
- Eggs are laid in water
- Larvae bears external gills



Diversity of Amphibians

Order: Urodela/Caudata

Reproduction

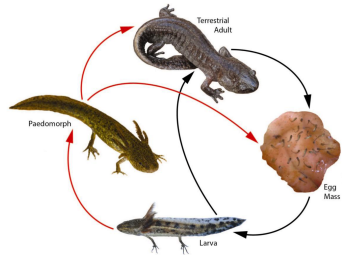
- Some are direct developing- bypass the aquatic larval stage, directly develop to miniature adults



Diversity of Amphibians

Order: Urodela/Caudata

- **Paedomorphosis: retention of ancestral juvenile features in later stages of development**
- Sexually mature larval morph of *Ambystoma* sp.- **Axolotl**



Diversity of Amphibians

Order: Anura/Salienta

- Anura (Gr. *an* - without, *oura* – tail)
- **Adults lack a tail**
- Old group known from the Jurassic period, 150 MYA
- More than 7150 species
- Commonly called **frogs and toads**
- **Specialized for jumping**
- Size
 - largest - Goliath frog >30cm
 - smallest - *Paedophryne amauensis* 7.7mm
- Usually found near water

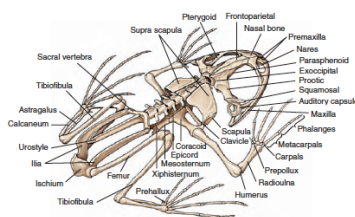


Diversity of Amphibians

Order: Anura/Salienta

Skeleton

- **Nine trunk vertebrae**
- **Rod like urostyle** – several fused caudal vertebrae
- Skull is light



Diversity of Amphibians

Order: Anura/Salienta

Locomotion

- Jumping, hopping, crawling, climbing, gliding, swimming
- **Webbed feet** – swimming
- **Tibia and fibula have been fused into a single, strong bone**
- **Radius and ulna in the fore limbs have fused into a single, strong bone**
- **Long limbs and sticky expanded toe pads – arboreal life**

Diversity of Amphibians

Order: Anura/Salienta

Respiration

- Use lungs, mouth and skin
- Anurans depend more lung respiration
- **Positive-pressure breather** - fills its lungs by forcing air into them
- Use rhythmic throat movements



Diversity of Amphibians

Order: Anura/Salienta

Vocalization

- Both male and female anurans have vocal cords
- Much better developed in males
- Males vocalize to attract females
- Produces sound by passing air back and forth over the vocal cords between the lungs and a large pair of sacs (vocal pouches) in the floor of the mouth



Diversity of Amphibians

Order: Anura/Salienta

Feeding

- Carnivorous, feed on insects, and anything that moves and is small enough to swallow whole
- Catch moving prey with the **protrusible tongue**
- Tongue is attached to the front of the mouth
- **Tongue is glandular and produces sticky secretions**
- Digestive tract is short in adults
- Larvae are mostly herbivorous, have a long digestive tract

